Install Manual
Full Build

How to install, update and look after the Node Controller web tools on a Raspberry Pi 3, 4 or 5 running AllStarLink with DVSwitch.

DASHBOARD SYSMON INSTMON WIFIMON 44HELPER

FILES COVERED BY THIS EDITION

TOOL	FILE	VERSION
instmon	instmon_v2_47_0_20261013.py	2.47.0
Installer	install_all_v7_1_20261012.sh	7.1
Dashboard	nodecontrol_v9_3_80_20261013.py	9.3.80
SysMon	sysmon_v6_13_85_20261014.py	6.13.85
wifimon	wifimon_v5_35_0_20261018.py	5.35.0
44helper	44helper_v0_0_207_20261018.py	0.0.207

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How to read the commands

Commands are typed on the Pi, in an SSH session or on a keyboard and screen. Lines starting with `#` are notes, not commands. Replace `<pi-address>` with the Pi's IP address or host name.

```
# a note about the next line
sudo systemctl status sysmon
```

NOTE The scripts on GitHub are comment-stripped copies. Each tool's change history is in the `changelogs/` folder of the repo.

NOTE 44helper has its own manual, tab by tab: `docs/ASL-DVS_44helper_Manual.pdf`. The user manual covers every tool: `docs/ASL-DVS_User_Manual.pdf`.

What gets installed

Each tool is a single file. Each one runs as its own systemd service and serves its own page in the browser.

TOOL	SERVICE	PORT	WHAT IT DOES
Dashboard	`nodecontrol`	8989	Node control. Connect ASL and EchoLink nodes; tune DMR, YSF, P25, NXDN, D-STAR, XLX and M17; favorites; Phone.
SysMon	`sysmon`	9999	System monitor and health checks for the node, DVSwitch and its config files.
instmon	`instmon`	8990	Installer and manager. Install, update, start and stop the other tools; GitHub updates; Quiet System.
wifimon	`wifimon`	8991	WiFi and supply-voltage watchdog. Shuts the Pi down cleanly on sustained low voltage or lost network.
44helper	`44helper`	9997	Optional. 44Net Connect, firewall, router and SvxLink helper.

The full build

SysMon and the Dashboard come in two builds. The full build is for a Pi 3, 4 or 5. Its files have no `_pi02w_` in the name, and its `VERSION` has no `-pi02w` ending.

DASHBOARD · FULL

Every tab, including Phone and all digital modes.

- Keeps the AllStar and EchoLink lists in memory for the fastest search.
- Polls Asterisk and refreshes the page at the full rate.

SYSMON · FULL

Every tab, including the ones the Pi02w build drops:

- Console: the in-browser Asterisk console and the Troubleshooting checks.
- Net, Reg, Firewall and Zello.

instmon, wifimon, 44helper and the installer are common: one build for every Pi.

PI ZERO 2 W? Use the Pi02w build and its own manual, `docs/ASL-DVS_Pi02w_Install_Manual.pdf`. The full build is too heavy for 512 MB.

NOTE The Watchdog script (`asl_dvs_watchdog`) is not needed. The Dashboard's own service already restarts it if it stops, and the installer removes an old Watchdog if it finds one.

Before you start

What the Pi needs

- A Raspberry Pi 3, 4 or 5 already running AllStarLink with DVSwitch (Analog_Bridge and MMDVM_Bridge) set up.
- `python3` and `systemd`. Both come with Raspberry Pi OS.
- A network connection, and SSH access with `sudo`.

Set a root password

The web pages ask for a login, and they check it against the Pi's root password. Raspberry Pi OS ships with root locked, so set a password once:

```
sudo passwd root
```

Get the files and install instmon

Get the files

Put all the files in one folder on the Pi, for example your home folder:

```
# files from the main branch on GitHub
cd ~
B=https://raw.githubusercontent.com/kd8pgk/ASL-DVS/main
wget $B/instmon_v2_47_0_20261013.py
wget $B/install_all_v7_1_20261012.sh
wget $B/nodecontrol_v9_3_80_20261013.py
wget $B/sysmon_v6_13_85_20261014.py
wget $B/wifimon_v5_35_0_20261018.py
# optional
wget $B/44helper_v0_0_207_20261018.py
```

TIP Newer files may be on GitHub by the time you read this. Check the repo's file list and use the newest SysMon and Dashboard files without `_pi02w_` in the name.

instmon goes first. Once it is running, it can install and update everything else from the browser, and it updates itself.

STEP 1 · INSTMON

```
sudo python3 instmon_v2_47_0_20261013.py --install
```

Then open `http://<pi-address>:8990` and log in as root.

What --install does

- Copies the file into place and writes `instmon.service`.
- Writes the shared launcher `/usr/local/bin/asl_dvs_launch.py` (see How the tools start, section 8).
- Enables the service so it starts on boot, and starts it now.

Optional: stage the other files in instmon

Instead of running the installer by hand (Step 2), you can upload the files on instmon's page, or let **Check GitHub** download them into instmon's library, and press **Install** there. The installer also finds files staged in instmon's library.

Install the rest with the installer

STEP 2 · INSTALLER

```
sudo bash install_all_v7_1_20261012.sh
```

What it asks

1. **Operation.** If tools are already installed: `i` install / reinstall, `u` update binaries only, or `q` quit. Config is kept either way.
2. **Version.** If more than one version of a tool is found, pick one (Enter takes the newest).
3. **First-run identity.** On a fresh node: your callsign, ASL node number and a display label. They go into `/etc/asl_dvs/asl_dvs.conf`.

Where it looks for files

The installer's own folder, the current folder, `/root`, `/home/pi`, your home folder, `/tmp` and instmon's library. Any subset of the tools can be installed: it installs what it finds.

Which build it picks

For SysMon and the Dashboard the build is chosen first, then only that build's files are offered:

1. `SYSMON_VARIANT` / `DASH_VARIANT` = `pi02w` or `full`, if set;
2. otherwise the build already installed;
3. otherwise `full` on a Pi 3, 4 or 5 (`pi02w` only on a Pi Zero 2 W).

```
# force the full build, for example to switch back from pi02w
sudo SYSMON_VARIANT=full DASH_VARIANT=full bash install_all_v7_1_20261012.sh
```

CAREFUL If no file of the chosen build is found, that tool is skipped with a warning. The other build is never used in its place.

Without questions

Add `--non-interactive` (or `-y`). It picks the newest files, updates an existing install, and takes the identity from environment variables on a fresh node:

```
sudo AUTO_CALLSIGN=KD8PGK AUTO_NODE=652701 AUTO_LABEL="Travel Node" \
     bash install_all_v7_1_20261012.sh --non-interactive
```

When it finishes, the installer checks that each service is running and listening on its port.

5 · STEP 3

Open the pages and log in

PAGE	ADDRESS
Dashboard	http://<pi-address>:8989
SysMon	http://<pi-address>:9999
instmon	http://<pi-address>:8990
wifimon	http://<pi-address>:8991
44helper	http://<pi-address>:9997

Log in as root with the root password from section 2. The pages use plain HTTP, so only open them on your own network or over a VPN.

> **TIP** On the first visit to the Dashboard, check that the callsign and node number in the header are right.

6 · STEP 4

Set up wifimon

On a first install, wifimon runs with both watchdogs off: it shows the WiFi and the voltage, but doesn't reconnect or shut the node down. When you log in to its page (port 8991) it asks "Turn the watchdogs on?". Add your WiFi networks first, then turn them on there or in the Watchdog card:

- **WiFi watchdog:** reconnects to saved networks, and shuts the node down after 3 minutes with no connection. It waits, with no timer, until at least one network is saved.
- **Voltage watchdog:** shuts the node down after 30 seconds of low voltage.

Updates and reinstalls keep the watchdogs as they were. Add the networks it may reconnect to:

```
sudo python3 /usr/local/bin/wifimon.py --setup-wifi
```

You can also manage networks and each WiFi adapter on wifimon's page (port 8991).

> **CAREFUL** wifimon shuts the Pi down cleanly on sustained low voltage or a sustained loss of network. Use a good power supply and cable: a Raspberry Pi is sensitive to voltage drop.

Updating from GitHub with instmon

The **GitHub Updates** card in instmon reads the repo's file list directly.

CHECK GITHUB

Shows the newest GitHub copy of each tool and whether it is newer than yours. Full and Pi02w builds get separate rows; rows for the build this Pi isn't using are tagged **other build**.

UPDATE

Downloads one file into instmon's library and checks it first: the same file GitHub listed, a version line, the right first line, a syntax check and `--install` support. Nothing installs until you press **Install**.

FULL UPDATE

1. Downloads and checks every new file before anything changes.
2. Pauses the web tools.
3. One tool at a time: saves a copy of the installed file, uninstalls it, installs the new one. If the new one fails, it puts the old one back and moves on.
4. Updates instmon last. A timer reinstalls the old instmon if the new one isn't answering after 90 seconds.
5. Ends with a summary.

Builds and Full Update

On a Pi 3, 4 or 5 each tool keeps the build it has, and Check GitHub only offers newer files of that build. The Pi Zero 2 W build choice only appears on a Pi Zero 2 W.

Quiet System

Quiet System / Restore pauses SysMon, the Dashboard and 44helper before a big upload or install. Asterisk, the bridges, Allmon3, SSH and wifimon keep running. Everything comes back by itself after 30 minutes.

How the tools start

TOOL	STARTED AS
Dashboard (full)	`/usr/bin/python3 /usr/local/bin/nodecontrol.py`
SysMon (full)	`/usr/bin/python3 /usr/local/bin/sysmon.py`
instmon, wifimon, 44helper	`/usr/bin/python3 /usr/local/bin/asl_dvs_launch.py <tool>`

The launcher

The common tools start through `/usr/local/bin/asl_dvs_launch.py`. Started directly, Python compiles the whole file on every start and keeps that memory. The launcher loads the tool as a module instead, so Python saves the compiled copy in `__pycache__` and reuses it. That is about half the memory, and twice as fast to start. The full SysMon and Dashboard are started directly.

Three more memory savers

Launcher version 2 also turns these on for the tools it starts. Each has an off switch:

SAVER	WHAT IT DOES	OFF SWITCH
-OO	Drops the built-in help text (docstrings) from the loaded code.	`/etc/asl_dvs/launch_no_optimize`
Pool cap	`MALLOC_ARENA_MAX=2`: at most 2 memory pools instead of up to 8 per CPU core.	`/etc/asl_dvs/launch_no_arena_cap`
Trim	1 minute after start, then every 5 minutes, hands freed memory back to Linux.	`/etc/asl_dvs/launch_no_trim`

To turn one off, create its file and restart the services. Delete the file and restart to turn it back on:

```
sudo touch /etc/asl_dvs/launch_no_arena_cap
sudo systemctl restart instmon wifimon 44helper
```

NOTE The first start after an update compiles the tool once more, so its memory runs high until the next restart.

Checking memory

```
free -m
# memory of each Python tool, in MB
ps -eo rss,args --sort=-rss | awk '/python3/ && !/awk/ {printf "%6.1f MB  %s\n", $1/1024, substr($0, index($0,$2))}'
```

Troubleshooting

A page doesn't load

```
# is the service running?
systemctl status sysmon --no-pager
# its last 30 log lines
journalctl -u sysmon -n 30 --no-pager
# is anything listening on the port?
ss -tlnp | grep 9999
```

Use the service name and port of the tool in question (table in section 1).

The login is refused

The pages check the root password. Set or reset it with `sudo passwd root`.

The installer says "No full ... build found"

The full-build file for that tool (no `_pi02w_` in the name) isn't in any of the folders it searches. Put it next to the installer and run it again.

The Pi is slow or runs out of memory

- Run the memory commands in section 8.
- Check that the instmon, wifimon and 44helper lines show `asl_dvs_launch.py`. Started as plain `python3 tool.py`, a tool uses about twice the memory: run its `--install` again.
- Check that swap is on: `swapon --show` should list it. If an older 44helper turned it off, use 44helper's System Optimization tab (**Swap -- restore**), or `sudo systemctl enable --now dphys-swapfile`.
- Close Cockpit's page when you're done: its two `cockpit-bridge` processes use about 27 MB while it is open.

A tool misbehaves after an update

Turn the memory savers off one at a time (section 8), starting with the pool cap, and restart. If that doesn't help, reinstall the previous version from instmon's library.

SysMon's own checks

```
sudo python3 /usr/local/bin/sysmon.py --selftest
```

It runs SysMon's security checks against this Pi and prints PASS / WARN / FAIL for each, without starting the web page.

On the page, **Console → Troubleshooting** runs the full set of checks, including the installed Dashboard file, the saved favorites and, from 6.13.84, the **Half tunnel** group.

10 · REMOVING

Removing the tools

Remove one tool with its own uninstall:

```
sudo python3 /usr/local/bin/sysmon.py --uninstall
```

Remove every tool except instmon:

```
sudo bash uninstall_all_v1_5.sh
```

This leaves `/etc/asl_dvs/`, instmon's library and `/etc/wifimon/wifimon.conf` in place. Remove instmon last with its own `--uninstall`. The launcher is removed once no service uses it.

11 · QUICK REFERENCE

Quick reference

TASK	COMMAND
Install instmon	`sudo python3 instmon_v2_47_0_20261013.py --install`
Install the rest	`sudo bash install_all_v7_1_20261012.sh`
Set the login password	`sudo passwd root`
Add WiFi networks	`sudo python3 /usr/local/bin/wifimon.py --setup-wifi`
Restart the web tools	`sudo systemctl restart sysmon nodecontrol instmon wifimon`
A tool's log	`journalctl -u <service> -n 30 --no-pager`
Memory overview	`free -m · zramctl`
Saver off switch	`sudo touch /etc/asl_dvs/launch_no_<optimize|arena_cap|trim>`
Remove all but instmon	`sudo bash uninstall_all_v1_5.sh`

SERVICE	PORT	CONFIG
nodecontrol	8989	/etc/asl_dvs/asl_dvs.conf
sysmon	9999	/etc/sysmon/sysmon.conf
instmon	8990	/etc/asl_dvs/ · library in /etc/asl_dvs/instmon_library
wifimon	8991	/etc/wifimon/wifimon.conf
44helper	9997	/etc/44helper/44helper.conf